Network Pruning for Remote Sensing Images Classification Based on Interpretable CNNs

Xianpeng Guo, *Student Member, IEEE*, Biao Hou, *Member, IEEE*, Bo Ren, *Member, IEEE*, Zhongle Ren, *Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Convolutional neural network (CNN)-based research has been successfully applied in remote sensing image classification due to its powerful feature representation ability. However, these high-capacity networks bring heavy inference costs and are easily overparameterized, especially for the deep CNNs pretrained on natural image datasets. Network pruning is regarded as a prevalent approach for compressing networks, but most existing research ignores model interpretability while formulating pruning criterion. To address these issues, a network pruning method for remote sensing image classification based on interpretable CNNs is proposed. More specifically, an original interpretable CNN with a predefined pruning ratio is trained at first. The filters, namely channels in the high convolutional layer, are able to learn specific semantic meanings in proportion to the predefined pruning ratio. The filters without interpretability are supposed to be removed. As for other convolutional layers, a sensitivity function is designed to assess the risk of pruning channels for each layer, and furthermore, the pruning ratio for each layer is corrected adaptively. The pruning method based on the proposed sensitivity function is effective and requires little computational costs to search abandoned channels without damaging classification performance. To demonstrate the effectiveness, the proposed method is implemented on different scales of modern CNN models, including VGG-VD and AlexNet. The experimental results, obtained on the UC Merced dataset and NWPU-RESISC45 dataset, prove that our method significantly reduces the inference costs and improves the interpretability of networks.**

***Index Terms*— Convolutional neural networks (CNNs), interpretable CNNs, network pruning, remote sensing image classification, sensitivity function.**

I. Introduction

In RECENT years, a wide range of scenarios, such as environment monitoring, urban planning, spatial-temporal data analysis, and smart city, raise an essential demand for automatic identification and classification of land use scenes. Consequently, the automation of the classification of remote sensing images has become a necessity.

Due to its broad application, there have been remarkable efforts in developing a variety of methods for remote sensing image classification during the past decades. The early methods generally devoted to extract a holistic and discriminative feature representation, such as bag-of-visual words (BoVW) [1], histogram of oriented gradient (HOG) [2], and scale-invariant feature transform (SIFT) [3]. However, heavily depending on the experience of researchers is an obvious drawback of feature extraction. Besides, with the constantly improving spatial resolutions in recent years, traditional frameworks emerged limitation of feature representation due to the great variations in the spatial arrangements and structural patterns.

More recently, by virtue of strong feature representation power in various vision tasks as well as powerful graphical processing units, deep learning algorithms, especially convolutional neural networks (CNNs), have been successfully applied to remote sensing image classification. Compared with the handcrafted feature-based methods, deep learning architecture was able to learn more abstract and discriminative semantic features with multiple convolutional layers, and thus, superior classification performance was achieved. The existent deep learning-based scene classification approaches can be concluded as three categories by different utilization of deep learning architecture [4]: training new networks [5]–[7], using pretrained networks [8]–[11], and making a pretrained model adapt [12]. Luus *et al.* [5], Zhang *et al.* [6], and Volpi and Tuia [7] constructed the networks whose architectures were relatively smaller compared with distinguished pretrained networks and trained them using satellite images from scratch. However, the generalization ability decreased on account of small-scale networks being subject to training datasets. Inspired by the prosperous application of pretrained CNN architectures such as AlexNet [13], VGG-VD [14], and GoogLeNet [15], some researchers directly extracted the features from intermediate layers of pretrained networks to form global feature representations [8]–[11]. Nonetheless, extracting intermediate layers and constructing reasonable combinations are still experience-dependent and verbose. Fine-tuning pretrained networks hence became a compendious and efficient alternative. AlexNet, VGG-16 [14], and so on were fine-tuned using remote sensing datasets in [12], obtaining more superior performance than using pretrained networks as feature extractors. An accompanying issue is that, the scales of public remote sensing images datasets are generally smaller than the

Manuscript received December 13, 2020; revised April 2, 2021; accepted April 27, 2021. Date of publication May 17, 2021; date of current version December 29, 2021. This work was supported in part by the Key Scientific Technological Innovation Research Project by Ministry of Education; in part by the National Natural Science Foundation of China under Grant 61671350, Grant 61771379, and Grant 61836009; in part by the Foundation for Innovative Research Groups of the National Natural Science Foundation of China under Grant 61621005; and in part by the Key Research and Development Program in Shaanxi Province of China (111 Project) under Grant 2019ZDLGY03-05. *(Corresponding author: Biao Hou.)*

The authors are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, International Research Center for Intelligent Perception and Computation, Joint International Research Laboratory of Intelligent Perception and Computation, Xidian University, Xi'an 710071, China (e-mail: guoxp@stu.xidian.edu.cn; avcodec@hotmail.com; asdwer2046@126.com; zlren@stu.xidian.edu.cn; lchjiao@mail.xidian.edu.cn).

Digital Object Identifier 10.1109/TGRS.2021.3077062

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.

natural images dataset consisting of millions of samples, which leads to a mass of redundancy on parameters and network structure. More than that, overparameterization takes high computational cost and high memory footprint for inference, which restricts the application of remote sensing techniques utilizing deep models on terminals with limited hardware.

To reduce the high computational cost, a widely recognized shortage of deep neural networks (DNNs), a bunch of network compression methods [16]–[23] from various perspectives have been proposed in recent years. Among all these methods, network pruning is identified as one of the most effective and prevalent approaches. Some researchers concentrated on individual weight pruning for networks. Pruning network weights with small magnitude was proposed by [24], and this method was further optimized to deeply compress networks in [25]. Louizos *et al.* [26] learned sparse networks during training by encouraging weights to become exactly zero. However, weight pruning methods result in unstructured sparse filters, which cannot bring compression and acceleration with frequently used hardware. Contrarily, benefited by the criteria that are adaptively computed with regard to the networks and needlessness of dedicated hardware, channel pruning methods have been widely adopted in a variety of scenarios [27]–[31]. He *et al.* [28] pruned each layer by a LASSO regression-based channel selection and least-square reconstruction. An automatic channel pruning method that used meta learning to choose pruned channels was presented in [31]. A max response selection strategy that pruned a certain percentage of filters with smaller kernel weights was proposed in [27]. This method significantly reduced computational cost in convolutional layers. Nevertheless, traditional max response selection strategy observed the sensitivity of each layer by gradually pruning filters and empirically determined whether prune layers or not, giving rise to additional computational cost as well as deviation caused by subjectivity. Moreover, an argument that deep layers are harder to prune than shallow layers was proposed [28], which may well be caused by larger receptive field and more powerful nonlinear feature description in the deep layers. Generally, filters in shallow layers describe simple textures and filters in deep layers are more likely to represent object parts [32]. As for remote sensing images, filters in high conv-layers generally represent important objects, such as airplanes, ships, highway, and bridge, which plays a decisive role in practical applications. Arbitrarily removing the filters, namely, channels in deep layers without considering the actual representation may lead to irretrievable loss, whereas predominant current channel pruning methods mainly determine the contribution of different channels by statistic metric or rule learning. We consequently figure to prune channels in deep layers of remote sensing image classification networks while reserving the filters with clarified knowledge representations.

Concepts of visual interpretability for deep learning are adopted in this article. Although CNNs have achieved a notable momentum in classification task, the end-to-end learning strategy brings about infrequent interpretability. The interpretability usually describes how human could understand the model mechanisms and predictions [33]. It can help ensure impartiality in decision-making and provides a truthful causality in model inference [34]–[37]. More recent years, an increasing number of researchers have developed models with interpretable knowledge representations to unfold the black box of CNNs [38]–[43]. Propagating gradients of feature maps to estimate the image regions was proposed in [38], [39]. An explanatory graph was learned to reveal the semantic hierarchy hidden inside pretrained CNNs [40]. Some researchers focused on learning networks with disentangled representations. Interpretable regions with CNN features (R-CNN) [41] learned qualitatively interpretable models for object detection without any part annotations for supervision. Utilizing an iterative routing-by-agreement mechanism to assign higher weights with low-layer capsules, capsule networks that encoded specific semantic concepts were presented in [42]. InfoGan [43] learned disentangled representations by maximizing the mutual information between a small subset of the latent variables and the observation, whereas these methods usually required ingenious design or unique mechanism. A method that modifies traditional CNNs into interpretable CNNs was presented in [44]. This method clarified knowledge representations of CNNs by adding loss for the output feature maps of filters in deep layers and hence could be applied in most typical CNNs used for classification task.

With the aforementioned consideration, this article aims to compress networks for remote sensing image classification and, in the meantime, improve the interpretability of the pruned networks. For this to occur, we put forward a channel pruning method based on interpretable CNNs. To be specific, given a predefined pruning ratio first, the initial overparameterized models are trained by the framework of interpretable CNNs. This learning process, just like traditional CNNs, is an end-to-end pattern and the training data are also the same as ordinary CNNs without the need for extra annotations. The filters in high convolutional layer could be able to learn specific semantic meanings in proportion to the pruning ratio during the training process. Consequently, the interpretability of filters provides a pruning metric for high convolutional layers that are considered as hard to prune. The filters without clarified knowledge representations are supposed to be removed. Furthermore, a channel pruning framework based on a max response selection strategy is proposed. We design a sensitivity function to assess the risk of pruning channels for each layer. The predefined pruning ratio will be corrected for different layers. Unlike the traditional way and some other channel pruning methods that are time-consuming and subjective, the proposed framework requires little computational costs to pick out removed channels without damaging classification performance. Experiments on the proposed method indicate that networks pruned by our approach are compressed to a great extent with a puny loss of accuracy, and simultaneously, the unpruned filters in high conv-layer possess explicit knowledge representations.

The main contributions of this article are as follows.

1) We propose a network pruning algorithm for remote sensing image classification. This algorithm achieves model compression and inference acceleration by removing redundant channels in convolutional layers, which provides CNNs, especially DNNs, more

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.

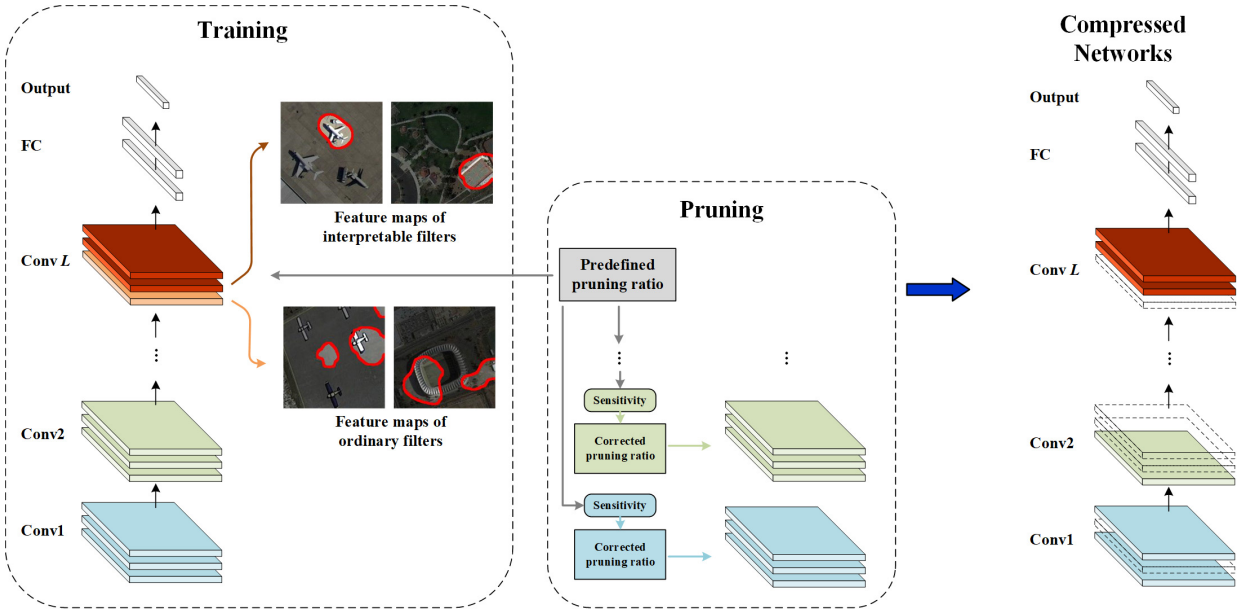


Fig. 1. Framework of the proposed method.

possibility to be applied in remote sensing applications with limited hardware resources.

2) The proposed network pruning method is based on interpretable CNNs. The compressed models are more semantically meaningful. This is beneficial for human to understand logic inside CNNs used for remote sensing image classification.
3) We propose a sensitivity function to evaluate the risk of pruning channels for each convolutional layer. The pruning ratio of each layer can be corrected adaptively and compression results are consequently improved.

The remainder of this article is organized as follows. In Section II, we detailedly introduce the proposed method. A series of experiments on four prevalent networks and two distinct datasets is reported in Section III. Comprehensive evaluations confirm the effectiveness of our proposed method. Finally, the conclusion and future work are drawn in Section IV.

II. Method

The integrated framework of the proposed method is shown in Fig. 1. Generally, a typical procedure of network pruning consists of three stages: 1) train an overparameterized model; 2) prune the trained large model according to a certain criterion; and 3) fine-tune the pruned model to regain the lost performance. We basically follow the strategy and our framework can be roughly divided into three procedures. As shown in Fig. 1, first, train an interpretable CNN with a predefined pruning ratio for remote sensing image classification. Filters in the last conv-layer are endowed with explicit knowledge representations by the predefined ratio, and furthermore, the filters without interpretability can be removed. During the pruning period, except for last conv-layer, prune the whole network by our proposed max response selection strategy based on the corresponding filters that have high absolute weights sum. The sensitivity of each conv-layer is obtained by the proposed sensitivity function, and then, the pruning ratio of each layer could be adaptively adjusted according to respective sensitivity. With that, every layer abandons a specific number of channels by the corrected pruning ratio. Finally, a fine-ili-tune operation is implemented on the compressed network to regain the lost performance.

A. Interpretable Convolutional Neural Networks

The interpretable CNNs framework was first proposed by Zhang *et al.* [44]. They proposed to slightly revise a CNN to improve its interpretability, with no need for any annotations of object parts or textures. Given a high conv-layer in a CNN, an extra loss for each filter in the conv-layer was added to push the filter toward the representation of an object part and the progress was end-to-end learning. Filters with extra loss could be exclusively activated by a certain category and keep silent on other categories. By virtue of this property, the filters in high conv-layer could be trained with interpretability by a predefined ratio.

Specifically, a loss is added on the feature map x of the filter f after the rectified linear unit (ReLU) operation in the target conv-layer. The feature map x after ReLU-layer is an $n \times n$ matrix, $x_{i,j} \geq 0$. Then, n^2 templates $\{T_{\mu_1}, T_{\mu_2}, \ldots, T_{\mu_{n^2}}\}$ are assigned to f since f's corresponding object part may appear at different locations in different images. Every template T_{μ_i} is also an $n \times n$ matrix. When a specific object part mostly activates the ith unit in feature map x, T_{μ_i} then describes the ideal distribution of activations for the feature map x.

During the forward propagation, when an image I is imported, a specific template $T_{\widehat{\mu}}$ from the n^2 templates candidates is selected as a mask to filter out noisy activations from x. Here, $\widehat{\mu} = \arg\max_{[i,j]} x_{ij}$ and $x^{\text{masked}} = \max\{x \circ T_{\widehat{\mu}}, 0\}$, where $\circ$ denotes the Hadamard (element-wise) operator. $\mu = [i, j]$, $1 \leq i, j \leq n$, denotes the location of potential object part in feature map x.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.

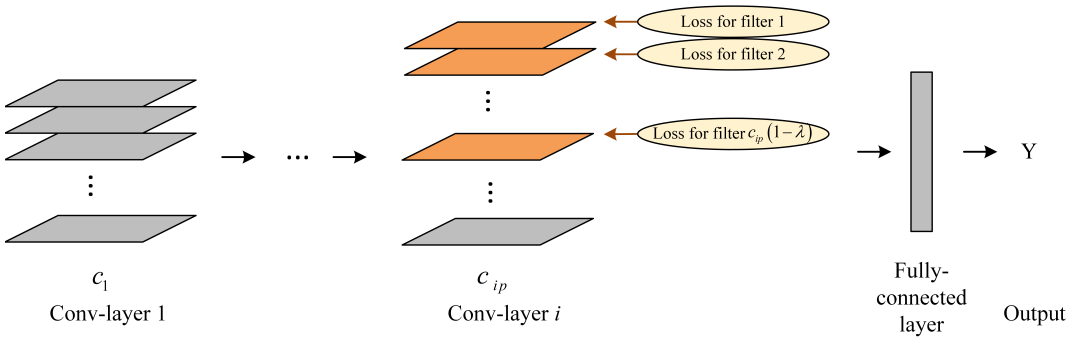


Fig. 2. Structure of transforming ordinary CNNs into interpretable CNNs. c_1 and c_{ip} indicate the channel number of conv-layer 1 and interpretable conv-layer, respectively. Gray and orange squares correspond to ordinary filters and interpretable filters, respectively.

The operation mentioned above supports the gradient backpropagation algorithm for end-to-end learning. When it comes to the backpropagation process, the added loss pushes filter f to represent a specific object part of a corresponding category c and keep inactivated on images of other categories. Let $\boldsymbol{X} = \{x|x = f(I)\}$ denote the feature maps of f after an ReLU operation and $\boldsymbol{T} = \{T^-, T_{\mu_1}, T_{\mu 2}, \ldots, T_{\mu_{n^2}}\}$ denotes template candidates. The loss for f is formulated by the mutual information between $\boldsymbol{X}$ and $\boldsymbol{T}$

$$\begin{aligned}\text{Loss}_f &= -MI(\boldsymbol{X}; \boldsymbol{T}) \\ &= -\sum_T p(T) \sum_x p(x|T) \log \frac{p(x|T)}{p(x)}. \end{aligned} \tag{1}$$

The prior probability of a template is defined as $p(T_\mu) = \frac{\theta}{n^2}$, $p(T^-) = 1 - \theta$, where θ is a constant prior likelihood. The fitness between a feature map x and a template T is measured as the conditional likelihood $p(x|T)$

$$\forall T \in \boldsymbol{T},\ p(x|T) = \frac{1}{Z_T} \exp[\text{tr}(x \cdot T)] \tag{2}$$

where $\cdot$ indicates the matrix multiplication, $Z_T = \sum_{x \in \boldsymbol{X}} \exp(\text{tr}(x \cdot T))$, $\text{tr}(\cdot)$ indicates the trace of a matrix, and $\text{tr}(x \cdot T) = \sum_{i,j} x_{ij} t_{ij}$, $p(x) = \sum_T p(T) p(x|T)$.

During the training period, if the input image $I \in I_c$, the feature map $x = f(I)$ is expected to the assigned template $T_{\widehat{\mu}}$; if $I \notin I_c$, the feature map x is supposed to match a designed negative template T^-. Especially, the negative template is omitted during the forward propagation. All feature maps will be assigned positive templates as masks.

The training process of interpretable CNNs is the same as traditional CNNs. In the forward-propagation period, each filter in the CNNs transmits its information in a bottom-up manner, and in the backpropagation period, each filter in an interpretable conv-layer receives gradients with respect to its feature map x from the final classification loss $L(\widehat{y}_k, y_k^*)$ as well as the local filter loss Loss_f. Hence, the gradients of filters in interpretable conv-layer are calculated as follows:

$$\frac{\partial Loss}{\partial x_{ij}} = \varphi \frac{\partial \text{Loss}_f}{\partial x_{ij}} + \frac{1}{N} \sum_{i=k}^{N} \frac{\partial L\left(\widehat{y}_k, y_k^*\right)}{\partial x_{ij}} \tag{3}$$

where φ is a weight.

B. Rectified Interpretable Convolutional Neural Networks

In original method, the interpretable conv-layer is added as an individual layer after the end conv-layer in the networks, which augment the scale of networks and amount of parameters. This is contrary to this article's motivation, and thus, we rectify the interpretable CNNs slightly. Loss_f is added to the filters in the end conv-layer, transforming the end conv-layer into interpretable conv-layer. The structure is shown schematically in Fig. 2. To prune channels in the interpretable conv-layer, the filters in interpretable conv-layer are divided into two kinds, interpretable ones and other without interpretability. Let λ denote the predefined pruning ratio that is manually set as a desired whole network compressed ratio and c_{ip} denote the number of channels in the interpretable layer, namely the number of filters. $c_{ip}(1-\lambda)$ filters are designed to add Loss_f, while the rest of filters, namely $c_{ip}\lambda$ filters, are trained in the traditional way. Therefore, the gradients of filters in interpretable conv-layer are changed to

$$\frac{\partial Loss}{\partial x_{ij}} = \varphi \beta_i \frac{\partial \text{Loss}_f}{\partial x_{ij}} + \frac{1}{N} \sum_{i=k}^{N} \frac{\partial L\left(\widehat{y}_k, y_k^*\right)}{\partial x_{ij}} \tag{4}$$

where $\beta \in \mathbb{R}^{c_{ip}}$ is a coefficient vector of length c_{ip} for channel pruning and β_i is a scalar mask to the ith filter. Notice that, when the pruning ratio λ is given, β is defined. $\beta_i = 1$, $1 \le i \le c_{ip}(1-\lambda)$ and $\beta_i = 0$, $c_{ip}(1-\lambda) \le i \le c_{ip}$. If $\beta_i = 0$, Loss_f will not be added to the filter.

When the interpretable CNNs are well trained, as mentioned in the previous subsection, $c_{ip}(1-\lambda)$ filters are supposed to learn explicit semantic representations, and these filters are protected from pruning. $c_{ip}\lambda$ filters without interpretability are removed away in the last conv-layer. Furthermore, when filters are pruned, the interpretable conv-layer with fewer filters is created and the remaining parameters of the unaffected filters are copied into the new conv-layer. The kernels in the next conv-layer corresponding to the pruned feature maps are also removed.

C. Max Response Selection Strategy With Sensitivity Function

The end conv-layer in the network has been pruned as mentioned above, and the remaining conv-layers are pruned by our proposed max response selection strategy based on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.

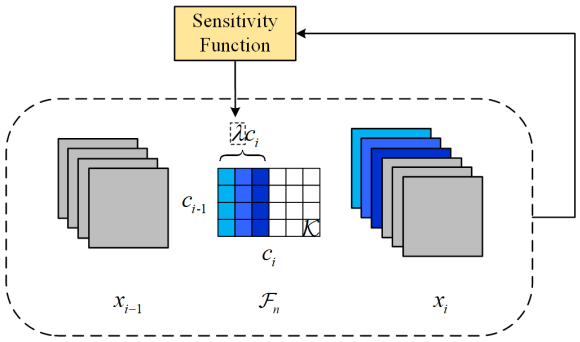


Fig. 3. Procedure of pruning channels. x_{i-1} and x_i represent feature maps of two consecutive convolutional layers. $\mathcal{F}_{i,j}$ is the kernel matrix. c_{i-1} and c_i represent the corresponding channel numbers. The blue squares represent pruning filters with relatively small absolute weights.

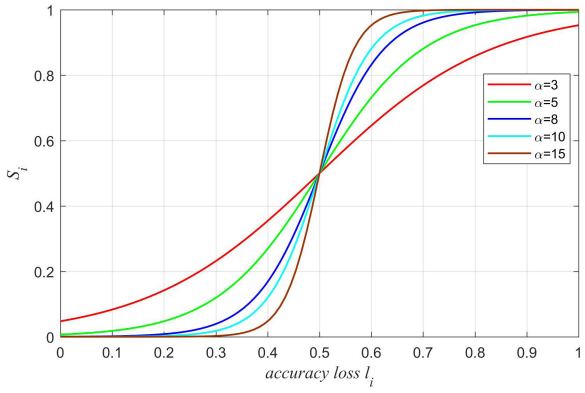


Fig. 4. Sensitivity function S_i under different values of α.

corresponding filters that have high absolute weights sum. It has been proved in [27] that if pruning channels in a specific conv-layer led to more accuracy loss than pruning other layers, namely this layer was more sensitive, reserving all channels of the sensitive layers achieved better compression performance. As a consequence, this article proposes a sensitivity function to quantize the sensitivity of each layer. The sensitive conv-layers are supposed to be protected from splitting or reserve channels as many as possible.

To be specific, let c_i denote the channel numbers, namely filter numbers in the ith conv-layer, and h_i, w_i be the height and width of the output feature maps. The output feature maps $x_i \in \mathbb{R}^{c_i \times h_i \times w_i}$ are achieved by applying c_i filters $\mathcal{F}_{m,n} \in \mathbb{R}^{c_{i-1} \times k \times k}$ on the c_{i-1} input channels, in which one filter generates one feature map. k represents the kernel size. Each filter is composed by c_{i-1} kernels $\mathcal{K}_{m,n} \in \mathbb{R}^{k \times k}$. The kernel matrix $\mathcal{F}_n \in \mathbb{R}^{c_{i-1} \times c_i \times k \times k}$ consists of all the filters. The relative importance of a filter in each layer is calculated by the sum of its absolute weights $\sum |\mathcal{F}_{m,n}|$, $\sum |\mathcal{F}_{m,n}|$ also represents the average magnitude of input channels since the number of input channels c_i is the same across filters. The absolute weights can be regarded as an expectation of the magnitude of the output feature maps. In the same conv-layer, filters with smaller kernel weights generate lower activations in feature maps compared with filters with larger kernel weights. Feature maps with lower activations play a less prominent part in classifier. Therefore, the filters with smallest kernel weights are supposed to be removed away, i.e., given a predefined pruning ratio λ, $c_i\lambda$ filters with the smallest kernel weights are removed in the ith layer.

Definitely, pruning every conv-layer by a fixed ratio λ as mentioned above is excessively arbitrary and ignoring the sensitivity of different layers. The traditional method observed sensitivity by comparing the accuracy loss between different layers after removing the same number of filters gradually. We further propose a sensitivity function to measure the sensitivity of each layer rather than judging by experience. The procedure is shown in Fig. 3. As shown in Fig. 3, the predefined pruning ratio λ is corrected by the algorithm via one pruning operation. For the ith conv-layer, $c_i\lambda$ filters with the smallest kernel weights are pruned first, and the accuracy loss l_i on the validation set can be achieved. Here, l_i means the difference of classification accuracy between unpruned networks and pruned networks. The sensitivity function S_i is formulated as follows:

$$S_i = \frac{1}{1 + \exp\left(\alpha \frac{l_{\max} + l_{\min} - 2l_i}{l_{\max} - l_{\min}}\right)} \tag{5}$$

where α is a positive constant to control the slope of S_i. $l_{\max}$ and $l_{\min}$, respectively, refer to the maximum accuracy loss and the minimum accuracy loss in l. Because the layer with maximum accuracy loss and those layers whose accuracy loss close to maximum are crucial for classification, they are expected to reserved or pruned as little as possible and vice versa. On account of this property, the sensitivity function S_i is designed as a variant of sigmoid function that is a nonlinear classifier. Noted that $S_i \in [(1/1 + e^{\alpha}), (1/1 + e^{-\alpha})]$, when l_i approaches maximum accuracy loss, S_i will approximate 1 by adjusting α within a range. Also, S_i is able to tend to 0 with l_i decreasing. Fig. 4 briefly shows the sensitivity function S_i under different constants α. It can be seen that when α approaches 1, the curve will close to a straight line, and the sensitivities of layers with accuracy loss that is close to maximum decline more. These layers will also be pruned. This is contrary to expectation. Hence, α is supposed to be a larger positive constant, such as 5–15.

The sensitivity function S_i describes importance of each convolutional layer for classification. Apparently, the corrected pruning ratio should to be scaled down if S_i is large, and when $S_i = 0$, the corrected pruning ratio should remain unchanged. Thus, the corrected pruning ratio for the ith conv-layer λ_i^{cor} is expected as

$$\begin{aligned} \lambda_i^{\text{cor}} &= \lambda(1 - S_i) \\ &= \lambda\left(1 - \frac{1}{1 + \exp\left(\alpha \frac{l_{\max} + l_{\min} - 2l_i}{l_{\max} - l_{\min}}\right)}\right) \end{aligned} \tag{6}$$

where $\lambda_i^{\text{cor}} \in (0, \lambda)$. λ_i^{cor} will close to zero when $l_i = l_{\max}$, and hence, the sensitive layers will be reserved from pruning. Also, if $l_i = l_{\min}$, λ_i^{cor} will keep the initial predefined pruning ratio. After that, the pruning number of filters for the ith conv-layer should be calculated as $c_i\lambda_i^{\text{cor}}$. The pruning

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.

number of different conv-layers varies with the sensitivity corresponding to specific conv-layer, rather than pruning the whole layers or part layers with the same pruning ratio, which shows the adaptivity to various conv-layers. The pseudocode of max response selection strategy with sensitivity function is shown in Algorithm. 1

Algorithm 1 Pseudocode of Max Response Selection Strategy With Sensitivity Function

Input: predefined pruning ratio λ, conv-layer L, channel numbers c
Output: corrected pruning ratio λ^{cor}

for $i = 1, \ldots, L$ **do**
  for $n = 1, \ldots, c_i$ **do**
    calculate $\sum \left|\mathcal{F}_{m,n}\right|$
  end for
  sort filters by $\sum \left|\mathcal{F}_{m,n}\right|$
  prune $c_i \lambda$ filters with smallest sum values
  calculate accuracy loss l_i
  update l
end for
for $i = 1, \ldots, L$ **do**
  compute sensitivity of conv-layer S_i using (5)
  compute corrected pruning ratio λ_i^{cor} using (6)
  update λ^{cor}
end for

D. Channel Pruning and Fine-Tuning

In summary, an interpretable CNN is first trained with a predefined pruning raio λ. After that, in the last conv-layer, $c_{ip}\lambda$ filters without interpretability are pruned. As for other conv-layers, $c_i \lambda_i^{\text{cor}}$ filters with the smallest sum values are removed from the ith layer. Furthermore, when channels are pruned, the conv-layers with fewer channels are created and the remaining parameters of the unaffected channels are copied into the new conv-layer. The kernels in the next conv-layer corresponding to the pruned feature maps are also removed. In this way, a compressed network with less parameters and inference costs is generated.

After removing channels from networks, the structural change of networks may temporarily cause performance degradation to some extent. Fine-tuning is a generally adopted way to compensate for accuracy loss in network pruning. In some cases, the compressed networks may achieve better performance after fine-tuning, due to a more reasonable architecture of pruned networks. The optimization settings of fine-tuning are the same as ones in the training period, but the epochs in which fine-tuning demands are relatively smaller. Besides, fine-tuning requires less run-time memory than original training, due to a narrower network.

III. Experimental Results

In this section, we evaluate the proposed method on two datasets designed for remote sensing image classification and four widely adopted CNNs with different scales. We first demonstrate the experiment setup, including introductions to datasets and networks, implementation details, and evaluation criteria for computation cost. In Sections III-B and III-C, we, respectively, report the performance of our algorithm on two datasets in detail. This article hammers at pruning networks for remote sensing image classification and reserving the interpretability of the pruned networks in the meanwhile, so the overall accuracy, pruning ratio, as well as interpretability analysis are reported to demonstrate the argument. Besides, a couple of experiments and remarkable structured pruning methods are implemented for comparison.

A. Experiment Setup

1) Datasets Description: Two publicly available datasets designed for remote sensing image classification with different quantities of images are adopted in our experiments. The first one is the UC Merced land-use dataset, which consists of images of 21 land-use scene categories selected from aerial orthoimagery [45]. Each class contains 100 images with the size of 256×256 pixels and one-foot spatial resolution. Fig. 5 shows some example images randomly selected from UCM. We implement data augmentation, including flipping and rotating on account of the limited amount of images. The other one is the NWPU-RESISC45 dataset that is extracted from Google Earth (Google Inc.) [46], covering more than 100 countries and regions all over the world. The NWPU-RESISC45 dataset contains 31 500 images, containing 45 scene classes with 700 images in each class. Every image is 256×256 pixels in the red-green-blue (RGB) color space. The spatial resolution varies from about 30 to 0.2 m per pixel for most of the scene classes. Compared with UCM, NWPU-RESISC45 not merely covers the most categories of UCM, such as airplane, baseball diamond, forest, freeway, intersection, and storage tank, but also includes more generalized scene classification data, such as commercial area, mountain, church, palace, cloud, and sea ice, making classification task and network pruning more challenging. Fig. 6 shows some example images from NWPU-RESISC45 with one sample per class, and these classes are chosen from the different ones compared with UCM.

2) Networks: To demonstrate the broad applicability of our method, four practical CNNs with different types of structures are tested with our method. They are AlexNet [13], VGG-S [14], VGG-16 [14], and VGG-19 [14]. The numbers of convolutional layers are 5, 5, 13, and 16, and there are three fully connected (FC) layers in all four networks. With the different types of structures, the computational cost varies considerably. Among four networks, the floating-point operations (FLOPs) of VGG-16 and VGG-19 are ten orders of magnitude, whereas the FLOPs of AlexNet and VGG-S are nine orders of magnitude. Meanwhile, VGG-19 takes the highest memory footprint for inference, with nearly six times the FLOPs of AlexNet. In addition, skipping connections in residual networks [47] usually leads to single feature map encoding patterns of different filters, namely rising the difficulty of interpretability, so residual networks are not discussed in this article.

3) Evaluation Criteria: In our experiment, classification performance is evaluated by overall accuracy and computa-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.

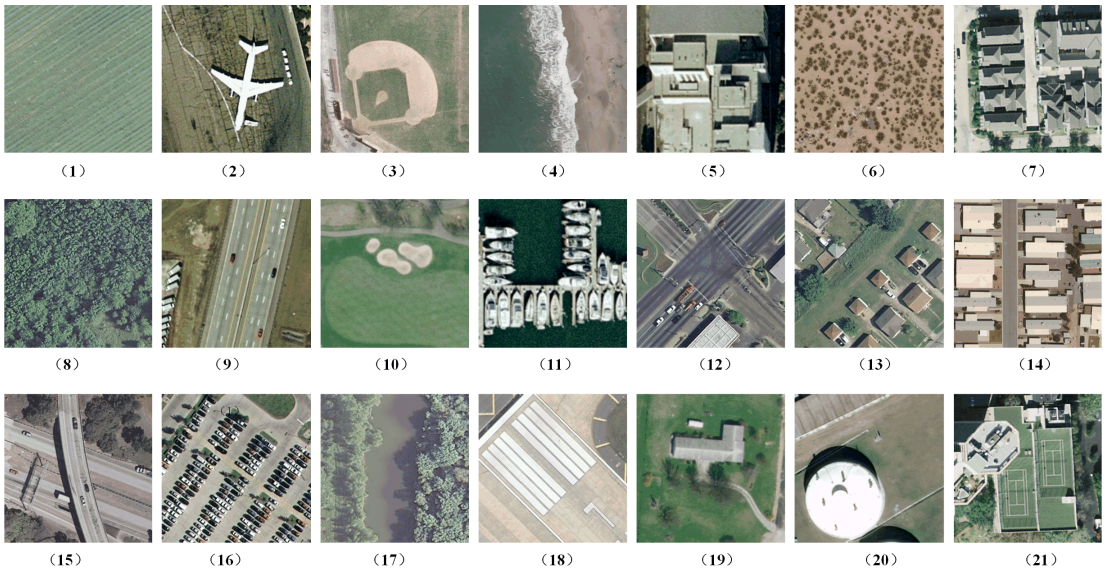


Fig. 5. Example images from the UCM dataset. (1) Agricultural. (2) Airplane. (3) Baseball diamond. (4) Beach. (5) Building. (6) Chaparral. (7) Dense residential. (8) Forest. (9) Freeway. (10) Golfcourse. (11) Harbor. (12) Intersection. (13) Medium residential. (14) Mobile home park. (15) Overpass. (16) Parking lot. (17) River. (18) Runway. (19) Sparse residential. (20) Storage tank. (21) Tennis court.



Fig. 6. Example images from the NWPU-RESISC45 dataset.

tional cost is measured by FLOPs. In CNNs, the vast majority of computational cost comes from convolutional layers. Thus, the FLOPs in existing articles generally only take count of convolutional layers and we also calculated FLOPs of networks in the same way. It is worth noting that we transform original networks into interpretable networks by adding a mask, whose function is similar to activation function during inference; for every filters in one specific convolutional layer, hence, this transformation increases FLOPs very little and could be ignored.

When it comes to interpretability analysis, we implement an intuitive way that visualizes the reserved filters in the last conv-layer. The method proposed by Zhou *et al.* [48] is adopted to compute receptive fields of neural activations of reserved interpretable filters, which are scaled up to the image resolution.

4) Comparison Experiments: In this section, we will introduce two experiments that are set up to keep our results comparable. At the same time, we compare two structured pruning methods that have achieved remarkable results on natural image benchmarks, such as CIFAR-10, CIFAR-100, and ImageNet.

a) Randomly pruning (RP): First, in order to evaluate the effectiveness of pruning filters with the smallest sum of absolute weights in uninterpretable conv-layers, we compare our algorithm with a naive channel pruning strategy, which prunes the channels randomly from every layer. For a fair comparison, the pruning channel number of each uninterpretable conv-layer is exactly the same as our algorithm and the identical channels of interpretable conv-layer are pruned.

b) Uniform pruning ratio (UR): Second, we implement a comparison experiment that prunes each uninterpretable

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.

conv-layer with a uniform predefined ratio to evaluate the proposed sensitivity function. Also, UR prunes the identical channels of interpretable conv-layer to our algorithm. During operation, we make FLOPs of UR approach ours as much as possible by finding out a proper predefined ratio and, in the meanwhile, guarantee that the UR's compression ratio of FLOPs is not more than ours; thereby, the proposed sensitivity function could be evaluated by comparing the classification accuracy.

c) $L1$-norm: The first structured pruning method is $L1$-norm-based filter pruning [27], as mentioned in Section I, and this method prunes a certain predefined percentage of filters with smaller $L1$-norm absolute weights and reduces computational cost for VGG-16 by up to 34% on CIFAR-10.

d) Network slimming (NS): The second method is NS that was proposed by Liu *et al.* [29]. NS imposes the $L1$-sparsity on channel-wise scaling factors during training and then prunes channels with lower scaling factors. Unlike the $L1$-norm-based filter pruning, NS is an automatic structured pruning method since the channel scaling factors are compared across layers and reduces computational cost for VGG-19 by up to 51% on CIFAR-10.

Noticed that, the first step of our algorithm is pretraining an original interpretable networks and prunes the networks afterward, whereas the $L1$-norm-based channel pruning and NS prune networks that are unmodified and pretrained with remote sensing image datasets. By this means, we evaluate the compression effectiveness as well as interpretability analysis of the proposed method.

5) Hardware and Software Environment: All the experiments are run on an HP Z8 workstation with two Intel Xeon Gold 5120 CPUs with 14 cores, two NVIDIA GeForce RTX 2080 Ti GPUs, and a 128-GB memory. As for development environment, we use MatConvNet [49], which is based on MATLAB.

B. Experiment on the UC Merced Image Dataset

We first employ our algorithm on the UCM dataset with 21 land-use scenes. The images in the UCM dataset are divided into subsets for training/fine-tuning, validation, and testing. In order to justify the feasibility of the proposed method, two different sets of ratios are taken. The first set is 60%, 20%, and 20%, and the second is 80%, 10%, and 10%. All the networks are trained using SGD [50], with a batch size of 64. The initial learning rate is set to 0.001 and is divided by 5, while the average value of the objective function over the validation set stops decreasing. We train and fine-tune networks for 20 and 10 epochs, respectively.

1) Analysis of Rectifying Interpretable Convolutional Neural Networks: We compare the performance and computational costs of the proposed rectified interpretable CNNs with original method [44] to analyze the effect of transforming the end conv-layer into interpretable conv-layer. Both methods are tested with pruning as well as without pruning. The results are shown in Table I, they are obtained under the training ratio of 80%, and the number behind name of networks represents predefined pruning ratio λ. As shown in the table, no matter without pruning or with pruning, the FLOPs and parameters of original interpretable CNNs are always larger than those of the proposed method. This is because the original method adds an extra conv-layer as the interpretable layer, which certainly enlarges the computational costs. Besides, the classification performance of the proposed method is much the same as the original method. Thus, transforming the end conv-layer into interpretable conv-layer is reasonable and economical as well.

2) Pruning Performance: Tables II and III present the detailed experimental results under two different training ratios. As shown in the tables, the overall accuracy, FLOPs, and parameter number of different types of unpruned networks are displayed at the top of each row. The next two lines in each row show the results of our algorithm. It can be seen that within tiny accuracy loss, all four networks are able to reduce the computational cost by pruning certain channels.

TABLE I
COMPARISON OF THE PROPOSED RECTIFIED INTERPRETABLE CNNS AND ORIGINAL METHOD ON THE UCM DATASET

Model	Method	Accuracy	FLOPs	Parameters
VGG-S	interpretable	97.42%	6.68E+09	1.01E+08
	our-interpretable	97.37%	5.99E+09	9.89E+07
	prune-interpretable-0.35	97.15%	4.05E+09	6.99E+07
	prune-ours-0.35	97.09%	3.76E+09	6.46E+07
AlexNet	interpretable	97.22%	4.25E+09	5.90E+07
	our-interpretable	97.86%	4.15E+09	5.84E+07
	prune-interpretable-0.45	96.96%	3.00E+09	3.92E+07
	prune-ours-0.45	97.14%	2.97E+09	3.90E+07
VGG-16	interpretable	98.04%	2.09E+10	1.37E+08
	our-interpretable	97.90%	2.04E+10	1.34E+08
	prune-interpretable-0.4	97.76%	1.14E+10	8.97E+07
	prune-ours-0.4	97.71%	1.08E+10	8.56E+07
VGG-19	interpretable	98.73%	2.50E+10	1.42E+08
	our-interpretable	98.93%	2.46E+10	1.40E+08
	prune-interpretable-0.4	98.59%	1.36E+10	8.92E+07
	prune-ours-0.4	98.76%	1.27E+10	8.83E+07

TABLE II
RESULTS ON THE UCM DATASET UNDER THE TRAINING RATIO OF 80%

Model	Accuracy	FLOPs	Pruned	Parameters	Pruned
VGG-S	97.38%	5.99E+09	—	9.89E+07	—
VGG-S-0.35	97.09%	3.76E+09	37.20%	6.46E+07	34.63%
AlexNet	96.90%	4.15E+09	—	5.84E+07	—
AlexNet-0.45	97.14%	2.97E+09	28.37%	3.90E+07	33.23%
VGG-16	97.92%	2.04E+10	—	1.34E+08	—
VGG-16-0.4	97.71%	1.08E+10	47.22%	8.56E+07	36.27%
VGG-19	99.29%	2.46E+10	—	1.40E+08	—
VGG-19-0.4	98.76%	1.27E+10	48.52%	8.83E+07	36.79%

TABLE III
RESULTS ON THE UCM DATASET UNDER THE TRAINING RATIO OF 60%

Model	Accuracy	FLOPs	Pruned	Parameters	Pruned
VGG-S	96.80%	5.99E+09	—	9.89E+07	—
VGG-S-0.35	96.67%	3.76E+09	37.20%	6.46E+07	34.63%
AlexNet	96.61%	4.15E+09	—	5.84E+07	—
AlexNet-0.45	96.79%	2.97E+09	28.37%	3.90E+07	33.23%
VGG-16	98.09%	2.04E+10	—	1.34E+08	—
VGG-16-0.4	97.87%	1.08E+10	47.22%	8.56E+07	36.27%
VGG-19	98.45%	2.46E+10	—	1.40E+08	—
VGG-19-0.4	98.52%	1.27E+10	48.52%	8.83E+07	36.79%

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.

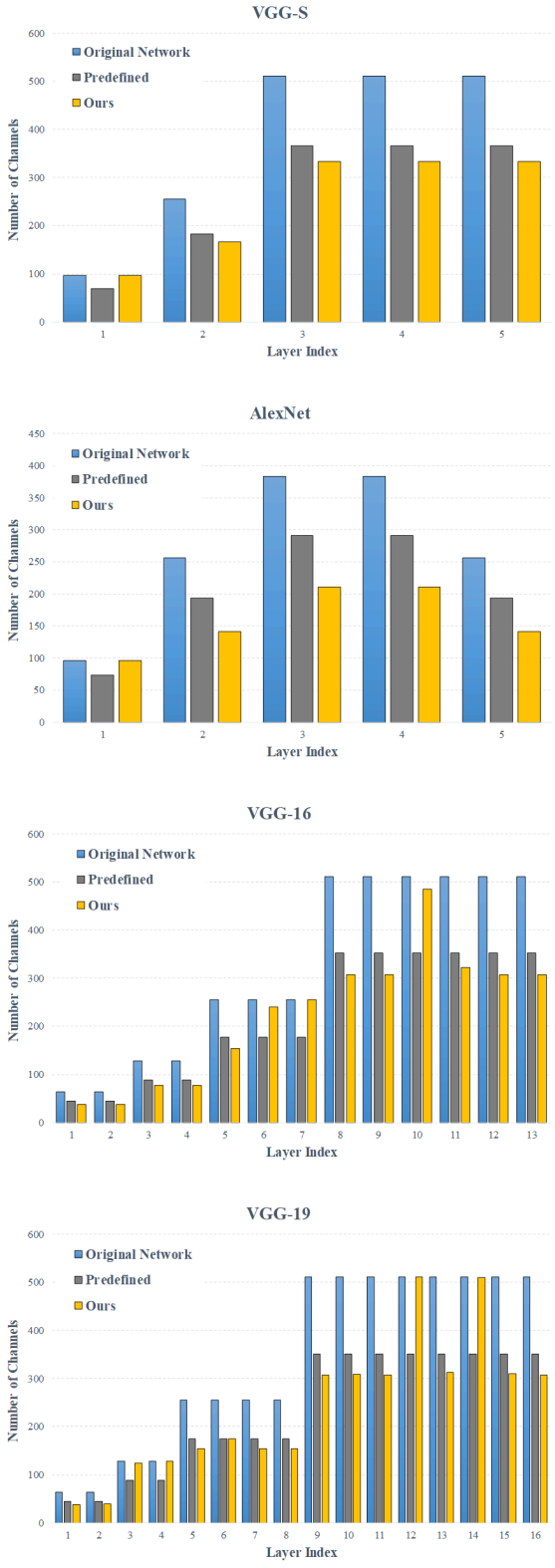


Fig. 7. Visualization of channel numbers of four networks on the UCM dataset. The blue bars represent the channels of unpruned networks. The gray bars represent the channels of networks pruned by a uniform predefined ratio method and the yellow bars represent the channels of networks pruned by our method.

Even we gain a higher overall accuracy than unpruned networks on AlexNet and VGG-19. One possible explanation we can deduce is that there is redundancy between network

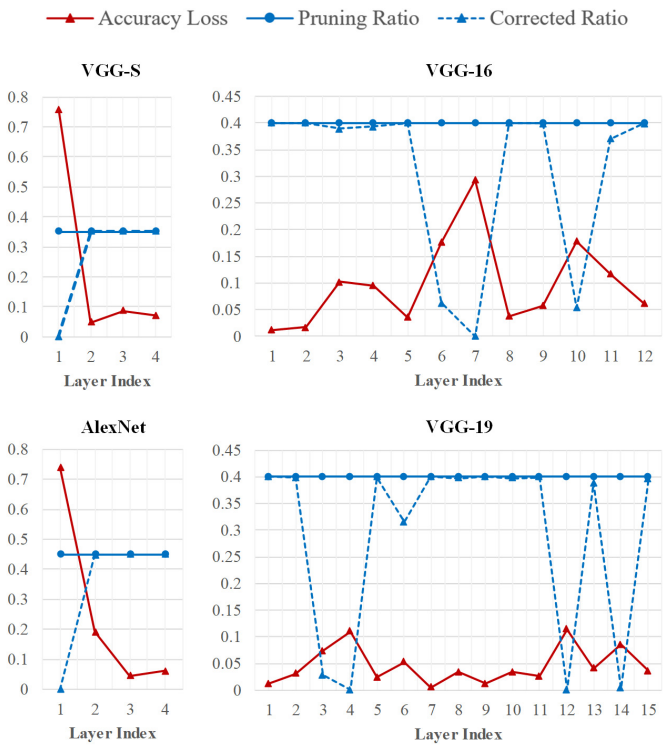


Fig. 8. Relationship of accuracy loss, predefined pruning ratio, and corrected pruning ratio.

TABLE IV
RELATIONSHIP OF ACCURACY LOSS, PREDEFINED PRUNING RATIO, AND CORRECTED PRUNING RATIO

	VGG-16-0.4		VGG-S-0.35	
Layer ID	Lost Accuracy	Corrected Ratio	Lost Accuracy	Corrected Ratio
1	1.19%	0.40	75.48%	0.00
2	1.67%	0.40	4.76%	0.35
3	10.24%	0.39	8.57%	0.35
4	9.52%	0.39	7.14%	0.35
5	3.57%	0.40		
6	17.62%	0.06		
7	29.29%	0.00		
8	3.81%	0.40		
9	5.71%	0.40		
10	17.86%	0.05		
11	11.67%	0.37		
12	6.19%	0.40		

connections, and thus, a more efficient network structure can be achieved by pruning superfluous channels. However, this benefit coexists with risk, and the accuracy could be irreparably injured if prune excess channels. On the other hand, we can preliminarily draw a conclusion that pruning channels on interpretable CNNs is feasible.

Fig. 7 shows the channel numbers of unpruned networks, models pruned by uniform predefined ratio method, and our pruned models (under training ratio of 80%). As shown in Fig. 7, our method safely prunes parts of channels in most convolutional layers with each network. However, there are several especial sensitive layers that are crucial to classification performance, and hence, these layers will not be pruned or pruned with seldom channels. In order to illustrate the mech-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.

TABLE V
CONSTRUCTION OF VGG-19 ON THE UCM DATASET. THE RIGHT TWO COLUMNS SHOW THE NUMBER OF CHANNELS AND REDUCED PERCENTAGE OF FLOPS

Layer Index	$w_i \times h_i$	Channels	FLOPs	Parameters	Channels	Pruned
Conv_1	224	64	8.99E+07	1.79E+03	38	40.63%
Conv_2	224	64	1.85E+09	3.69E+04	39	63.78%
Conv_3	112	128	9.26E+08	7.39E+04	124	40.90%
Conv_4	112	128	1.85E+09	1.48E+05	128	3.12%
Conv_5	56	256	9.26E+08	2.95E+05	154	39.84%
Conv_6	56	256	1.85E+09	5.90E+05	175	58.87%
Conv_7	56	256	1.85E+09	5.90E+05	154	58.87%
Conv_8	56	256	1.85E+09	5.90E+05	154	63.80%
Conv_9	28	512	9.25E+08	1.18E+06	307	63.92%
Conv_10	28	512	1.85E+09	2.36E+06	309	63.81%
Conv_11	28	512	1.85E+09	2.36E+06	308	63.69%
Conv_12	28	512	1.85E+09	2.36E+06	512	39.84%
Conv_13	14	512	4.63E+08	2.36E+06	313	38.87%
Conv_14	14	512	4.63E+08	2.36E+06	510	39.10%
Conv_15	14	512	4.63E+08	2.36E+06	310	39.69%
Conv_16	14	512	4.63E+08	2.36E+06	307	63.69%
FC_1	7	4096	5.04E+09	1.03E+08	4096	40.04%
FC_2	1	4096	1.68E+07	1.68E+07	4096	0.00%
FC_3	1	21	8.60E+04	8.60E+04	21	0.00%
Total			2.46E+10	1.40E+08		48.52%

TABLE VI
PERFORMANCE COMPARISON OF THE PROPOSED METHOD AND FOUR CHANNEL PRUNING METHODS ON THE UCM DATASET

Model	Method	Lost Accuracy	FLOPs	Pruned	Parameters	Pruned
VGG-S	Ours-0.35	0.29%	3.76E+09	37.20%	6.46E+07	34.63%
	RP	0.95%	—	—	—	—
	UR-0.29	2.86%	3.78E+09	37.01%	7.42E+07	24.99%
AlexNet	Ours-0.45	-0.24%	2.97E+09	28.37%	3.90E+07	33.23%
	RP	0.57%	—	—	—	—
	UR-0.24	1.04%	2.97E+09	28.28%	4.76E+07	18.38%
VGG-16	Ours-0.4	0.21%	1.08E+10	47.22%	8.56E+07	36.27%
	RP	0.63%	—	—	—	—
	UR-0.31	0.82%	1.08E+10	47.04%	9.47E+07	29.50%
	L1-norm	1.97%	1.26E+10	38.04%	7.35E+07	45.28%
VGG-19	Ours-0.4	0.52%	1.27E+10	48.52%	8.83E+07	36.79%
	RP	1.43%	—	—	—	—
	UR-0.32	1.95%	1.27E+10	48.47%	9.67E+07	30.74%
	NS	0.57%	1.20E+10	50.98%	6.84E+07	51.04%

anism of the proposed sensitivity function, we also display the accuracy loss l_i that is used to compute the sensitivity function, the predefined pruning ratio λ, and the corrected ratio λ_i^{cor} calculated by sensitivity function in Fig. 8 and Table IV. The red lines in Fig. 8 indicate the accuracy loss after pruning λ channels in each convolutional layer. It can be seen from Fig. 8 that if the accuracy loss of one layer is obviously larger than those of the others, the proposed method will push the pruning ratio approach zero and, then, the sensitive layers will be protected from pruning. For example, first conv-layers of AlexNet and VGG-S are definitely more sensitive than the other layers, and as for VGG-16 and VGG-19, a few of middle and high layers play more important role in classification performance.

Here, we also list the construction of VGG-19 on the UCM dataset and the pruned model in Table V so as to detailedly display the pruned result. $w_i \times h_i$ means the size of output feature maps and the last two columns show the number of channels and the reduced percentage of FLOPs from the pruned model. As shown in Table V, the computational costs mainly come from convolutional layers and the FC layers occupy little because of the small size of output feature maps. Although the parameters of FC layers account for a relatively large proportion, we only prune convolutional layers considering that inference efficiency is the uppermost criteria for network compressing.

Table VI shows our experimental results and comparisons (under training ratio of 80%), in which the lost accuracy stands by the difference of the accuracy between unpruned networks and pruned model. The pruning ratios are indicated behind every method. The L1-norm prunes a half number of channels in conv-layers 8–13. The hyperparameter that controls the tradeoff of NS is set as 10^{-4}. It can be seen from Table VI that the proposed method generally outperforms RP and UR, which verifies the effectiveness of pruning filters with the smallest sum of absolute weights and the proposed sensitivity function as well. Our algorithm drops less accuracy than $L1$-norm and reduces more FLOPs at the same time. The performance of the proposed method and NS is about the same on compression of FLOPs, but NS prunes more parameters. On the one hand, NS prunes not only convolutional layers but also FC layers. On the other hand, there is a possible inference

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.



Fig. 9. Visualization of filters in last conv-layers of compressed AlexNet and compressed VGG-16 on the UCM dataset. Images above the red dotted line display the receptive fields of reserved filters in our method. Images below the red dotted line display the receptive fields of reserved filters in the $L1$-norm and NS methods.

TABLE VII
RESULTS ON THE NWPU-RESISC45 DATASET UNDER THE TRAINING RATIO OF 80%

Model	Accuracy	FLOPs	Pruned	Parameters	Pruned
VGG-S	86.76%	5.99E+09	—	9.90E+07	—
VGG-S-0.25	86.60%	4.36E+09	27.29%	7.30E+07	26.27%
AlexNet	85.54%	4.15E+09	—	5.85E+07	—
AlexNet-0.2	85.44%	3.60E+09	13.22%	4.97E+07	14.98%
VGG-16	93.60%	2.04E+10	—	1.34E+08	—
VGG-16-0.3	93.37%	1.17E+10	42.77%	9.61E+07	28.51%
VGG-19	94.54%	2.46E+10	—	1.40E+08	—
VGG-19-0.3	94.41%	1.37E+10	44.20%	9.87E+07	29.37%

TABLE VIII
RESULTS ON THE NWPU-RESISC45 DATASET UNDER THE TRAINING RATIO OF 20%

Model	Accuracy	FLOPs	Pruned	Parameters	Pruned
VGG-S	72.72%	5.99E+09	—	9.90E+07	—
VGG-S-0.25	71.93%	4.36E+09	27.29%	7.30E+07	26.27%
AlexNet	75.44%	4.15E+09	—	5.85E+07	—
AlexNet-0.2	75.38%	3.60E+09	13.22%	4.97E+07	14.98%
VGG-16	89.22%	2.04E+10	—	1.34E+08	—
VGG-16-0.3	89.20%	1.19E+10	41.91%	9.63E+07	28.37%
VGG-19	90.28%	2.46E+10	—	1.40E+08	—
VGG-19-0.3	90.19%	1.42E+10	42.42%	9.96E+07	28.72%

that interpretable networks make the connections between layers more correlative, and thus, interpretable networks might be harder to prune compared with the normal networks.

3) Interpretability Analysis: The following is interpretability analysis of models pruned by the proposed method. Fig. 9 shows the receptive fields of reserved filters in the last conv-layers of pruned networks. The shown images are randomly chosen from images corresponding to feature maps with high activation scores and the unpruned channels corresponding to these feature maps are also randomly selected. Here, we take AlexNet and VGG-16 for examples, and in the next section, we will show examples of VGG-S and VGG-19 on the NWPU-RESISC45 dataset. For comparison, we display receptive fields of reserved filters pruned by $L1$-norm and NS. As shown in Fig. 9, the reserved filters, namely channels, in interpretable networks are able to learn discriminative target or part of target in specific scene, yet reserved filters in normal networks usually do not have explicit semantic meanings. For example, airplanes in scene of airplane, boats docked together in scene of harbor, and bunker in scene of golf course are precisely learned. There is another form that filters learn part of target and edge feature together, such as junction of roads in scene of overpass, storage tank, and its edge in scene of storage tank. However, it is hard to understand the receptive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.

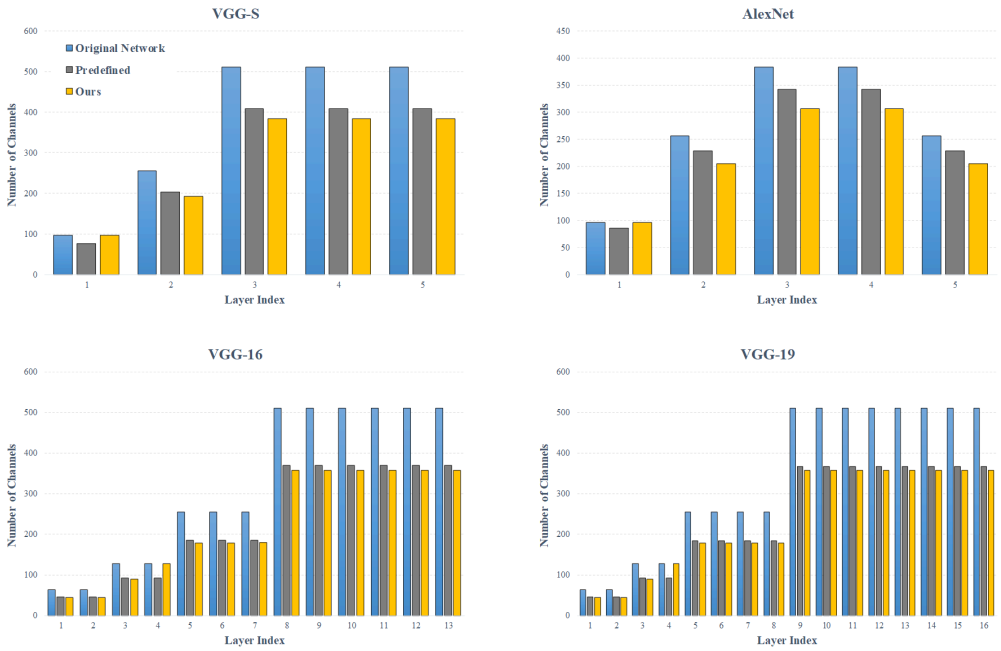


Fig. 10. Visualization of channel numbers of four networks on the NWPU-RESISC45 dataset. The blue bars represent the channels of unpruned networks. The gray bars represent the channels of networks pruned by a uniform predefined ratio method and the yellow bars represent the channels of networks pruned by our method.

TABLE IX
PERFORMANCE COMPARISON OF THE PROPOSED METHOD AND FOUR CHANNEL PRUNING METHODS ON THE NWPU-RESISC45 DATASET

Model	Method	Lost Accuracy	FLOPs	Pruned	Parameters	Pruned
VGG-S	Ours-0.25	0.16%	4.36E+09	27.29%	7.30E+07	26.27%
	RP	0.47%	—	—	—	—
	UR-0.2	1.85%	4.36E+09	27.25%	8.13E+07	17.90%
AlexNet	Ours-0.2	0.10%	3.60E+09	13.22%	4.97E+07	14.98%
	RP	0.35%	—	—	—	—
	UR-0.11	0.45%	3.61E+09	12.90%	5.36E+07	8.38%
VGG-16	Ours-0.3	0.23%	1.17E+10	42.77%	9.61E+07	28.51%
	RP	0.46%	—	—	—	—
	UR-0.28	0.73%	1.17E+10	42.68%	9.89E+07	26.43%
	L1-norm	1.47%	1.26E+10	38.04%	7.36E+07	45.25%
VGG-19	Ours-0.3	0.13%	1.37E+10	44.20%	9.87E+07	29.37%
	RP	0.67%	—	—	—	—
	UR-0.29	1.37%	1.37E+10	44.08%	1.01E+08	27.61%
	NS	0.11%	1.31E+10	46.58%	6.26E+07	55.18%

fields of reserved filters obtained by comparison methods. As can be seen from the above examples, CNNs pruned by the proposed method are able to keep excellent interpretability. The knowledge representations of these reserved filters in high conv-layers are not only helpful to understand logic inside a CNN but also coincidentally extract important target information in most remote sensing images.

C. Experiment on the NWPU-RESISC45 Dataset

Next, we implement our algorithm on the NWPU-RESISC45 dataset, which contains more categories and makes the task more challenging. Here, we also test two different training ratios to verify the feasibility of the proposed method. One set of ratios for training/fine-tuning, validation, and testing is 80%, 10%, and 10%, respectively. Another one is set as 20%, 10%, and 70%, which is the same as in [46]. All the networks are also trained using SGD, with a batch size of 64. The initial learning rate is set to 0.05 and is divided by 5, while the average value of the objective function over the validation set stops decreasing. We train and fine-tune networks for 40 and 10 epochs, respectively, on the NWPU-RESISC45 dataset.

1) Pruning Performance: The pruning results are shown in Tables VII and VIII. As shown in the tables, the classification accuracy under the training ratio of 20% is comprehensively lower than the results under the training ratio of 80%, but the pruning performance of the proposed method is not affected by the training ratio. Due to the increment of scene classes scale, the classification performance of each network generally declines compared with results on the MCU dataset. The difficulty of pruning task also increases, especially for networks with small scale such as VGG-S and AlexNet, and only 13.22% FLOPs are pruned for AlexNet. However, there are still many channels removed safely in networks with a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.



Fig. 11. Visualization of filters in the last conv-layers of compressed VGG-S and compressed VGG-19 on the NWPU-RESISC45 dataset. Images above the red dotted line display the receptive fields of reserved filters in our method. Images below the red dotted line display the receptive fields of reserved filters in the $L1$-norm and NS methods.

relatively large scale. This indicates that there might be more redundant connections with the increment of depth in CNNs.

Fig. 10 shows the channel numbers of four networks on the NWPU-RESISC45 dataset. Similarly, the results shown in the above two figures are all taken from experiments with a smaller pruning ratio on every networks. It can be seen that the proposed method prevents sensitive layers from pruning and generates more efficient networks. By contrast, UR removes numerous channels that are crucial to classification. There is a difference from results on the UCM dataset emerged in Fig. 10, the sensitive layers of large-scale networks mainly appear in the middle and lower layers.

Table IX summaries our experimental results with compared methods on the NWPU-RESISC45 dataset (under the training ratio of 80%). The performance of ours still outperforms RF and UR obviously. However, the compression ability is slightly inferior to NS. As for $L1$-norm, because more middle and higher layers that account for a large proportion of parameters are removed, it cuts down more percentage of parameters. Even so, we reduce more computational cost and drop much less accuracy than it.

2) Interpretability Analysis: In Fig. 11, the receptive fields of reserved filters in the last conv-layers of pruned AlexNet and VGG-19 networks are displayed. Here, we choose categories that the UCM dataset does not contain to visualize. As shown in Fig. 11, basketball court in scene of basketball court, parterre in scene of roundabout, stadium in scene of stadium, and cooling tower in scene of thermal power are clearly represented with our algorithm. Nevertheless, the receptive fields of reserved filters with conventional methods are jumbled and lack semantically meaningful.

IV. Conclusion

DNNs have been widely applied in remote sensing image classification due to its strong feature representation power. However, despite the remarkable discrimination power, the accompanying issues, such as high computational cost and black-box property, are always two bottleneck problems. To address these issues, this article proposes a heuristic channel pruning algorithm to compress models and advocates the use of training interpretable networks to give filters with clarified knowledge representations. A simple but effective sensitivity function is proposed to measure the contribution of each convolutional layer for classification. The sensitivity function enables the pruning ratio of each convolutional layer to be adjusted adaptively. Then, the filters in the last conv-layers with semantic meanings are reserved by training interpretable CNNs with a predefined pruning ratio. These filters of compressed networks represent discriminative object parts in specific scene.

Finally, this article carries out extensive experiments over two publicly available datasets designed for remote sensing image classification and four popular CNNs with different scales. Experimental results verify the feasibility of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.

proposed method. The effectiveness of the proposed sensitivity function is verified by comparative experiments. Compared with existing research, networks pruned by our method reveal more interpretability and are more helpful for human to understand.

In future work, we aim to explore methods to prune interpretable CNNs from scratch for remote sensing image classification. Pruning from scratch is able to save more computational cost because the pruning channels are already selected during training original overparameterized networks. Also, it is consistent with the procedure of pruning the last convolutional layer in this article's proposed method.

REFERENCES

[1] Q. Zhu, Y. Zhong, B. Zhao, G.-S. Xia, and L. Zhang, "Bag-of-visual-words scene classifier with local and global features for high spatial resolution remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 6, pp. 747–751, Jun. 2016.

[2] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, Jun. 2005, pp. 886–893.

[3] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.

[4] X. X. Zhu *et al.*, "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.

[5] F. P. S. Luus, B. P. Salmon, F. van den Bergh, and B. T. J. Maharaj, "Multiview deep learning for land-use classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 12, pp. 2448–2452, Dec. 2015.

[6] F. Zhang, B. Du, and L. Zhang, "Scene classification via a gradient boosting random convolutional network framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1793–1802, Mar. 2016.

[7] M. Volpi and D. Tuia, "Dense semantic labeling of subdecimeter resolution images with convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 881–893, Feb. 2017.

[8] O. A. B. Penatti, K. Nogueira, and J. A. dos Santos, "Do deep features generalize from everyday objects to remote sensing and aerial scenes domains?" in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2015, pp. 44–51.

[9] H. Fan, X. Gui-Song, H. Jingwen, and Z. Liangpei, "Transferring deep convolutional neural networks for the scene classification of high-resolution remote sensing imagery," *Remote Sens.*, vol. 7, no. 11, pp. 14680–14707, 2015.

[10] D. Marmanis, M. Datcu, T. Esch, and U. Stilla, "Deep learning Earth observation classification using ImageNet pretrained networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 1, pp. 105–109, Jan. 2016.

[11] G.-S. Xia *et al.*, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[12] K. Nogueira, O. A. B. Penatti, and J. A. dos Santos, "Towards better exploiting convolutional neural networks for remote sensing scene classification," *Pattern Recognit.*, vol. 61, pp. 539–556, Jan. 2017.

[13] A. Krizhevsky, I. Sutskever, and G. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1–9.

[14] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent.*, 2014, pp. 1–14.

[15] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.

[16] M. Jaderberg, A. Vedaldi, and A. Zisserman, "Speeding up convolutional neural networks with low rank expansions," in *Proc. Brit. Mach. Vis. Conf.*, 2014, pp. 1–12.

[17] X. Zhang, J. Zou, K. He, and J. Sun, "Accelerating very deep convolutional networks for classification and detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 10, pp. 1943–1955, Oct. 2016.

[18] M. Wang, B. Liu, and H. Foroosh, "Factorized convolutional neural networks," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2017, pp. 545–553.

[19] M. Danelljan, G. Bhat, F. S. Khan, and M. Felsberg, "ECO: Efficient convolution operators for tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6931–6939.

[20] B. Peng, W. Tan, Z. Li, S. Zhang, D. Xie, and S. Pu, "Extreme network compression via filter group approximation," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 300–316.

[21] M. Rastegari, V. Ordonez, J. Redmon, and A. Farhadi, "Xnor-net: Imagenet classification using binary convolutional neural networks," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 525–542.

[22] M. Courbariaux, Y. Bengio, and J.-P. David, "Binaryconnect: Training deep neural networks with binary weights during propagations," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 3123–3131.

[23] M. Courbariaux, I. Hubara, D. Soudry, R. El-Yaniv, and Y. Bengio, "Binarized neural networks: Training deep neural networks with weights and activations constrained to +1 or -1," 2016, *arXiv:1602.02830*. [Online]. Available: http://arxiv.org/abs/1602.02830

[24] S. Han, J. Pool, J. Tran, and W. Dally, "Learning both weights and connections for efficient neural network," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 1135–1143.

[25] S. Han, H. Mao, and W. J. Dally, "Deep compression: Compressing deep neural networks with pruning, trained quantization and Huffman coding," in *Proc. Int. Conf. Learn. Represent.*, 2015, pp. 1–14.

[26] C. Louizos, M. Welling, and D. P. Kingma, "Learning sparse neural networks through L_0 regularization," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–13.

[27] H. Li, A. Kadav, I. Durdanovic, H. Samet, and H. P. Graf, "Pruning filters for efficient convnets," in *Proc. Int. Conf. Learn. Represent.*, 2016, pp. 1–13.

[28] Y. He, X. Zhang, and J. Sun, "Channel pruning for accelerating very deep neural networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1389–1397.

[29] Z. Liu, J. Li, Z. Shen, G. Huang, S. Yan, and C. Zhang, "Learning efficient convolutional networks through network slimming," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2736–2744.

[30] J.-H. Luo, J. Wu, and W. Lin, "ThiNet: A filter level pruning method for deep neural network compression," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 5058–5066.

[31] Z. Liu *et al.*, "MetaPruning: Meta learning for automatic neural network channel pruning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3296–3305.

[32] D. Bau, B. Zhou, A. Khosla, A. Oliva, and A. Torralba, "Network dissection: Quantifying interpretability of deep visual representations," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6541–6549.

[33] P. Hall, "On the art and science of machine learning explanations," 2018, *arXiv:1810.02909*. [Online]. Available: http://arxiv.org/abs/1810.02909

[34] B. Goodman and S. Flaxman, "European union regulations on algorithmic decision-making and a 'right to explanation,'" *AI Mag.*, vol. 38, no. 3, pp. 50–57, Oct. 2017.

[35] F. Zhuang, X. Cheng, P. Luo, S. J. Pan, and Q. He, "Supervised representation learning: Transfer learning with deep autoencoders," in *Proc. 24th Int. J. Conf. Artif. Intell.*, 2015, pp. 1–7.

[36] D. Gunning, "Explainable artificial intelligence (XAI)," in *Proc. Defense Adv. Res. Projects Agency (DARPA) Web*, vol. 2, no. 2, 2017, pp. 1–18.

[37] A. Barredo Arrieta *et al.*, "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Inf. Fusion*, vol. 58, pp. 82–115, Jun. 2020.

[38] R. C. Fong and A. Vedaldi, "Interpretable explanations of black boxes by meaningful perturbation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 3429–3437.

[39] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.

[40] Q. Zhang, R. Cao, F. Shi, Y. N. Wu, and S.-C. Zhu, "Interpreting CNN knowledge via an explanatory graph," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 4454–4463.

[41] T. Wu, W. Sun, X. Li, X. Song, and B. Li, "Towards interpretable R-CNN by unfolding latent structures," 2017, *arXiv:1711.05226*. [Online]. Available: http://arxiv.org/abs/1711.05226

[42] S. Sabour, N. Frosst, and G. E. Hinton, "Dynamic routing between capsules," in *Proc. Adv. Neural Inf. Proces. Syst.*, 2017, pp. 3856–3866.

[43] X. Chen, Y. Duan, R. Houthooft, J. Schulman, I. Sutskever, and P. Abbeel, "Infogan: Interpretable representation learning by information maximizing generative adversarial nets," in *Proc. Adv. Neural Inf. Proces. Syst.*, 2016, pp. 2172–2180.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.

[44] Q. Zhang, Y. N. Wu, and S.-C. Zhu, "Interpretable convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8827–8836.
[45] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst. (GIS)*, 2010, pp. 270–279.
[46] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.
[47] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[48] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Object detectors emerge in deep scene CNNs," in *Proc. Int. Conf. Learn. Represent.*, 2015, pp. 1–12.
[49] A. Vedaldi and K. Lenc, "MatConvNet: Convolutional neural networks for MATLAB," in *Proc. 23rd ACM Int. Conf. Multimedia*, Oct. 2015, pp. 689–692.
[50] L. Bottou, "Stochastic gradient descent tricks," in *Neural Networks: Tricks of the Trade*, vol. 7700. Springer, 2012, pp. 421–436.



Xianpeng Guo (Student Member, IEEE) received the B.S. degree from Xidian University, Xi'an, China, in 2015, where he is pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China.

His research interests include deep learning and classification and object detection in remote sensing images.



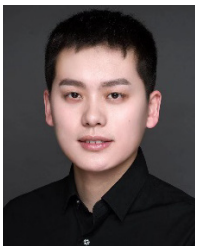
Biao Hou (Member, IEEE) received the B.S. and M.S. degrees in mathematics from Northwest University, Xi'an, China, in 1996 and 1999, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2003.

Since 2003, he has been with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University, where he is a Professor. His research interests include compressive sensing and synthetic aperture radar image interpretation.



Bo Ren (Member, IEEE) received the B.S. degree in telecommunications engineering from Northwest University, Xi'an, China, in 2011, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2017.

Since 2018, he has been an Assistant Professor with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China. He is a Visiting Scholar with the GIPSA-lab laboratory (Grenoble Images Parole Signal Automatique), Grenoble, France. His research interests include synthetic aperture radar image interpretation and understanding and sparse representation, and machine learning in remote sensing images.



Zhongle Ren (Member, IEEE) received the B.S. degree in electronic and information engineering and the Ph.D. degree in circuit and system from Xidian University, Xi'an, China, in 2013 and 2019, respectively.

He is a Lecturer with the School of Artificial Intelligence, Xidian University. His research interests include machine learning, deep learning, and classification and object detection in remote sensing images.



Licheng Jiao (Fellow, IEEE) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Electronic Engineering, Xidian University, Xi'an, where he is the Director of the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China and the International Research Center of Intelligent Perception and Computation. His research interests include intelligent information processing, image processing, machine learning, and pattern recognition.

Dr. Jiao is a member of the IEEE Xi'an Section Execution Committee, the President of the Computational Intelligence Chapter, the IEEE Xi'an Section, and the IET Xi'an Network, the Chairman of the Awards and Recognition Committee, the Vice Board Chairperson of the Chinese Association of Artificial Intelligence, a Councilor of the Chinese Institute of Electronics, a Committee Member of the Chinese Committee of Neural Networks, and an Expert of the Academic Degrees Committee of the State Council.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:33:27 UTC from IEEE Xplore. Restrictions apply.